

TEERTHA MAZUMDER

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PROFESSIONAL SUMMARY

Versatile technology professional with a Master of Applied Science in Software Engineering and broad expertise spanning enterprise infrastructure and systems administration, full-stack software development, AI and machine learning engineering, data analytics, and Microsoft ecosystem technologies. Currently serving as a Computer Systems Analyst II within the Government of Newfoundland and Labrador, supporting mission-critical enterprise systems including PeopleSoft HCM, Oracle Database, RHEL, and IBM AIX environments, and VMware virtualized infrastructure. Independently builds and ships production-grade AI-powered SaaS platforms, LLM agent systems, and cloud native applications. Proficient in Python, JavaScript, TypeScript, React, Node.js, SQL, Power BI, REST API development, LangChain, LangGraph, RAG pipelines, Docker, Kubernetes, and cloud platforms including AWS, Azure, and GCP. Recognized for analytical thinking, clear communication with both technical and non-technical stakeholders, and a strong commitment to delivering reliable, high-quality solutions in collaborative team environments.

TECHNICAL SKILLS

Systems Administration: Red Hat Enterprise Linux RHEL, IBM AIX, Windows Server, VMware vSphere, IBM Power Systems HMC and LPAR, Apache HTTPD and Tomcat, IBM Spectrum Protect, server installation, configuration, patching, monitoring, and troubleshooting

Microsoft Ecosystem: Power BI (advanced DAX, data modeling, dashboard development, report publishing), Microsoft Excel (advanced), Microsoft Office Suite, Azure App Services and Functions (familiar), Azure Key Vault (familiar), Power Platform and Copilot Studio (familiar), C# and .NET (working knowledge)

AI and Machine Learning: Python, LangChain, LangGraph, Ollama, LLaMA, Anthropic Claude API, Azure OpenAI compatible models, Voyage AI, Hugging Face, RAG pipelines, prompt engineering, tool calling, agent orchestration, MCP concepts, Scikit-learn, TensorFlow, PyTorch, Keras, NLP, deep learning, embeddings, pgvector

Full Stack Development: React, TypeScript, JavaScript ES6+, Next.js 14, Node.js, PHP, HTML5, CSS3, Tailwind CSS, REST API design and integration, WebSockets, JWT authentication, OAuth, role-based access control, component architecture

Data and Analytics: Power BI, Tableau, Pandas, NumPy, Matplotlib, SQL, Oracle Database, MySQL, PostgreSQL, SQLite, Supabase, pgvector, ETL pipelines, data cleaning, data modeling, KPI reporting

DevOps and Cloud: AWS, Azure, GCP, Docker, Kubernetes, Terraform, Ansible, GitHub Actions CI/CD, Git, Linux, secure coding practices, secrets management, automated testing, Agile, and Scrum

PROFESSIONAL EXPERIENCE

Computer Systems Analyst II,

July 2025 to April 2026

Office of the Chief Information Officer, OCIO, Government of Newfoundland and Labrador, St. John's, NL

- Provide technical analysis, systems support, and solution design for enterprise IT infrastructure serving government departments and agencies across the province, supporting mission-critical applications, including PeopleSoft HCM and Financial Management Systems on RHEL, IBM AIX, and VMware vSphere environments.
- Administer and optimize Oracle Database and MySQL environments, performing SQL query optimization, data integrity validation, performance tuning, and supporting analytical data flows for enterprise applications.
- Develop Python and Bash automation scripts for system monitoring, log-level analysis, statistical reporting, and alert delivery, building maintainable, well-documented production pipelines with scheduled CI/CD-style execution.
- Support enterprise system integrations between PeopleSoft HCM, government data systems, and external services, analyzing data flows, identifying mapping issues, and coordinating resolutions across application and infrastructure teams.
- Administer Apache HTTPD, Apache Tomcat, IBM Spectrum Protect, and IBM Power Systems HMC, and LPAR virtualized infrastructure, managing server lifecycle including installation, configuration, patching, and decommissioning.
- Produce technical documentation, solution designs, configuration standards, and operational runbooks aligned with enterprise architecture, security, and change management standards.
- Collaborate with cross-functional teams and communicate complex technical findings clearly to both technical and non-technical stakeholders, maintaining a client-focused, service-oriented approach.

- Provide technical guidance, peer review, and knowledge transfer to colleagues, contributing to team capability and a culture of continuous improvement.

Data Analyst and Systems Engineer Contract

2020 to 2023

PROSHIKA A Center for Human Development Dhaka, Bangladesh

- Designed and maintained end-to-end data processing and reporting systems using Python and SQL, managing data extraction, cleaning, transformation, validation, and delivery across organizational information systems.
- Built Power BI and Tableau dashboards to monitor KPIs, analyze program trends, and communicate performance insights clearly to executive stakeholders and non-technical audiences.
- Automated ETL workflows using Python and SQL scripting, reducing manual overhead and improving data consistency and accuracy across reporting pipelines.
- Collaborated with business teams to define KPIs, gather requirements, and iterate on data solutions aligned with organizational goals, applying strong analytical and stakeholder management skills.

KEY PROJECTS

BotNest Multi-Tenant SaaS AI Platform

2026

Trapezium Personal Venture

- Built a full-stack multi-tenant SaaS platform using React, TypeScript, Next.js 14, Supabase PostgreSQL with pgvector, Voyage AI embeddings, and the Anthropic Claude API, implementing a complete RAG pipeline with prompt engineering, grounding, tool calling, and streamed LLM responses deployed to production with CI/CD and live monitoring.
- Designed secure multi-tenant architecture with OAuth-compatible authentication, Row Level Security, rate limiting, secrets management, and Stripe Connect billing, applying enterprise-grade security and governance patterns.

OpenCare Two-Sided Healthcare Marketplace

2026

Trapezium Personal Venture

- Building a cloud native healthcare marketplace using Next.js 14, React, TypeScript, Supabase, Daily.co real-time video, and Stripe Connect, independently owning system design, API contracts, security, and cloud deployment for a regulated healthcare context requiring data integrity and reliability.

WinAgent Local Windows AI Automation Agent

2025 to 2026

Personal Project

- Built a production-ready AI agent using Python, LangChain, LangGraph, and Ollama, implementing multi-step agentic reasoning with LangGraph state machines for tool calling, branching task execution, and system automation, applying MCP-style agent tool integration patterns and evaluating multiple open-source LLMs for performance and trade-offs.

Electronic Design Automation Using LLMs | MASc Research

2024

Memorial University of Newfoundland, supervised by Dr. Lihong Zhang

- Designed Python-based data processing, orchestration, and evaluation pipelines using LLaMA to benchmark AI-assisted automation, applying prompt engineering, RAG patterns, performance testing, and clear technical communication of findings to diverse audiences.

Athlete Management and Real-Time Tracking Web Application

2024

Memorial University of Newfoundland supervised by Dr. Adrian Fiech

- Built a full-stack web application using Node.js, React, and RESTful APIs with JWT authentication, role-based access control, and live data integration between frontend and backend services.

Sentiment Analysis Engine and Knowledge Graph

2018 to 2019

Daffodil International University

- Built a deep learning NLP sentiment classification model using TensorFlow, Keras, and NLTK, and a knowledge graph using Python and NetworkX, demonstrating applied ML pipeline design, model evaluation, and data engineering.

Online Result Management System | BSc Research Project

2019

Daffodil International University | Academic

- Developed a role-based web platform for academic result publication using PHP, HTML/CSS, and MySQL for a national education board in Bangladesh.

E-Commerce Food Ordering System

2021

Daffodil International University | Commercial

- Developed an Android-based food ordering application for e-commerce operations. Used SQLite for storing and managing product, customer, and order-related data. Explored ASP.NET backend integration for order management and reporting. Improved understanding of mobile app development, database design, and backend connectivity.

Cost Optimized NAS and Linux Infrastructure

2020

Daffodil International University | commercial

- Designed and implemented a Linux based NAS server with Bash automation, RAID, LVM, SSH hardening, and storage management.

EDUCATION

Master of Applied Science MASc Software Engineering

2023 to 2024

Memorial University of Newfoundland, St. Johns NL

Research: Electronic Design Automation using Large Language Models, Python, LLaMA, NLP, RAG, prompt engineering, AI benchmarking, and evaluation

Bachelor of Science in Computer Science and Engineering

2016 to 2019

Daffodil International University, Dhaka, Bangladesh

CERTIFICATIONS

- Red Hat Certified System Administrator RHCSA Deploy, Configure, and Manage File Access, Storage, and Security
- Linux System Engineering and Security Networking, SSH Patch Management, Automation, CompTIA CEU eligible
- Learning Ansible Configuration Management
- Data Analyst and Development Certificate Excel Essential Training NASBA
- Emergency First Aid with CPR-B and CPR-C